

July 2026

The Editor-in-Chief
IEEE Internet of Things Journal

Dear Editor,

We are pleased to submit our manuscript entitled “Congestion-Aware Joint Charging-Station Placement, Capacity, and Swarm-UAV Trajectory Design for Age-Optimal Data Collection” for consideration for publication in the *IEEE Internet of Things Journal*.

Unmanned aerial vehicles are increasingly used to collect time-sensitive data from Internet of Things sensors, where the relevant quality metric is the Age of Information. Practical deployments use a swarm of energy-limited UAVs that must recharge at a small number of shared charging stations. This creates a problem that a single UAV never faces: the charging stations become a contended resource, and the resulting waiting time inflates the Age of Information. Deciding where to place the stations, how many charging ports to install, how the swarm should fly, and which UAV recharges where is a coupled design problem that, to our knowledge, has not been addressed as a whole.

The manuscript makes the following contributions:

- We formulate the first joint charging-station placement, port-capacity, swarm-trajectory, and assignment problem for peak-Age-of-Information data collection with a contended, multi-server charging resource.
- We show that the charging system must be modeled as a finite-source (machine-repair) queue rather than an open $M/M/c$ queue, and we validate this against a discrete-event simulation: the open queue overpredicts the waiting time by roughly six times and would distort the design.
- We prove two structural results: a congestion threshold that makes the Age of Information non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a placement inversion in which the optimal placement becomes traffic-driven rather than coverage-driven.
- We design a block-coordinate solver with a GPU close-enough traveling-salesman trajectory optimizer, and we evaluate it against single-UAV, single-station, coverage-optimal, and round-robin baselines, including a large-swarm scaling study, a robustness study, and an optimality-gap study against the exact optimum.

We believe the topic is well aligned with the scope of the *IEEE Internet of Things Journal* and will interest its readership working on UAV-assisted data collection, Age of Information, and energy-aware Internet of Things systems.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and its submission. We are not aware of any conflict of interest. To support reproducibility, the source code and data that generate all of our results are publicly available at <https://github.com/haodpsut/swarm-uav-aoi-congestion>.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Truong Duy Dinh (corresponding author),
on behalf of Phuc Hao Do, Tran Duc Le, Truong Duy Dinh, and Van Dai Pham
E-mail: duydt@ptit.edu.vn